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1.0 Specific Information:

No.	Unit No.	Serial Number	TC	Test Requested
1.	MAIN TRAFO UNIT 1	100691	4	Acidity, Dielectric, Color, Visualization and Water Content
2.	TRAFO 1 ESP UNIT 1	10937	5	Acidity, Dielectric, Color, Visualization and Water Content
3.	TRAFO 2 ESP UNIT 1	10035	6	Acidity, Dielectric, Color, Visualization and Water Content
4.	TRAFO 3 ESP UNIT 1	10031	7	Acidity, Dielectric, Color, Visualization and Water Content
5.	TRAFO 4 ESP UNIT 1	10034	8	Acidity, Dielectric, Color, Visualization and Water Content
6.	MAIN TRAFO UNIT 3	1122170	9	Acidity, Dielectric, Color, Visualization and Water Content
7.	UAT 3	1035082	10	Acidity, Dielectric, Color, Visualization and Water Content
8.	TRAFO 1 ESP UNIT 3	0073084208	11	Acidity, Dielectric, Color, Visualization and Water Content
9.	TRAFO 2 ESP UNIT 3	0070084208	12	Acidity, Dielectric, Color, Visualization and Water Content
10.	TRAFO 3 ESP UNIT 3	0072084208	13	Acidity, Dielectric, Color, Visualization and Water Content
11.	TRAFO 4 UNIT 3	0071084208	14	Acidity, Dielectric, Color, Visualization and Water Content
12.	ASH HANDLING	10820108	15	Acidity, Dielectric, Color, Visualization and Water Content
13.	ASH HANDLING	10820107	16	Acidity, Dielectric, Color, Visualization and Water Content

Note:

* OLTT+ consists of Liquid Screen Tests, Dissolved Gas Analysis (DGA), Water Content, Furan Analysis, Metals-in-Oil Analysis, Liquid Power Factors, and Inhibitor Content (DBPC).

** Liquid Screen Test consists of Acidity, Interfacial Tension (IFT), Dielectric, Colour, Specific Gravity, Visualization, and Sediment.

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2.0 Our Comments and Recommendations of The Test Results Analysis:

We had reviewed each laboratory test result against respective benchmark parameter and following are our comments and recommendations.

No.	Unit No.	Serial Number	TC
1.	MAIN TRAFO UNIT 1	100691	4

A) Analysis

- **Acidity** of 0.017mgKOH/g is **ACCEPTABLE**.
- **Dielectric breakdown voltage** of 92kV is **ACCEPTABLE**.
- **Colour** of 0.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 6ppm is **ACCEPTABLE** while moisture in solid insulation of 0.49% M/dw is graded as **A**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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No.	Unit No.	Serial Number	TC
2.	TRAFO 1 ESP UNIT 1	10937	5

A) Analysis

- **Acidity** of 0.056mgKOH/g is **QUESTIONABLE**. The level remained stable without further oil degradation.
- **Dielectric** of 65kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 18ppm is **ACCEPTABLE** while moisture in solid insulation of 1.89% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- It is still recommended to perform **Transformer Oil Regeneration** on this unit as per previous recommendation (unfavorable Acidity and IFT values). Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
3.	TRAFO 2 ESP UNIT 1	10035	6

A) Analysis

- **Acidity** of 0.042mgKOH/g is **ACCEPTABLE**. The level remained stable without further oil degradation.
- **Dielectric** of 42kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 14ppm is **ACCEPTABLE** while moisture in solid insulation of 1.65% M/dw is graded as **C**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- It is still recommended to perform **Transformer Oil Regeneration** on this unit as per previous recommendation (unfavorable IFT value). Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
4.	TRAFO 3 ESP UNIT 1	10031	7

A) Analysis

- **Acidity** of 0.042mgKOH/gm is **ACCEPTABLE**. The levels remained stable without further oil degradation.
- **Dielectric** of 61kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 13ppm is **ACCEPTABLE** while moisture in solid insulation of 1.37% M/dw is graded as **C**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
5.	TRAFO 4 ESP UNIT 1	10034	8

A) Analysis

- **Acidity** of 0.028mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 44kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 13ppm is **ACCEPTABLE** while moisture in solid insulation of 1.37% M/dw is graded as **C**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- It is still recommended to perform **Transformer Oil Regeneration** on this unit as per previous recommendation (unfavorable IFT and Liquid Power Factor values). Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
6.	MAIN TRAFO UNIT 3	1122170	9

A) Analysis

- **Acidity** of 0.017mgKOH/gm is **ACCEPTABLE**.
- **Dielectric** of 86kV is **ACCEPTABLE**.
- **Colour** of 0.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 8ppm is **ACCEPTABLE** while moisture in solid insulation of 0.57% M/dw is graded as **A**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
7.	UAT 3	1035082	10

A) Analysis

- **Acidity** of 0.022mgKOH/gm is **ACCEPTABLE**.
- **Dielectric** of 89kV is **ACCEPTABLE**.
- **Colour** of 0.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 11ppm is **ACCEPTABLE** while moisture in solid insulation of 0.78% M/dw is graded as **A**. For this transformer, it is recommended to maintain the water content less than 30ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
8.	TRAFO 1 ESP UNIT 3	0073084208	11

A) Analysis

- **Acidity** of 0.028mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 36kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 22ppm is **QUESTIONABLE** while moisture in solid insulation of 2.31% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content less than 20ppm. High level of moisture content will act as catalyst to the oxidation process.

B) Recommendation

- Based on the unfavorable IFT value from earlier oil analysis and current Water Content amount, it is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
9.	TRAFO 2 ESP UNIT 3	0070084208	12

A) Analysis

- **Acidity** of 0.109mgKOH/g is **UNACCEPTABLE**. The level remained stable without further oil degradation.
- **Dielectric** of 34kV is **ACCEPTABLE**.
- **Colour** of 2.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 17ppm is **ACCEPTABLE** while moisture in solid insulation of 2.21% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- It is still recommended to perform **Transformer Oil Regeneration** on this unit as per previous recommendation (unfavorable Acidity and IFT values). Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
10.	TRAFO 3 ESP UNIT 3	0072084208	13

A) Analysis

- **Acidity** of 0.025mgKOH/g is **ACCEPTABLE**.
- **Dielectric** of 50kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 21ppm is **QUESTIONABLE** while moisture in solid insulation of 1.49% M/dw is graded as **C**. For this transformer, it is recommended to maintain the water content less than 20ppm. High level of moisture content will act as catalyst to the oxidation process.

B) Recommendation

- Based on the unfavorable IFT value from earlier oil analysis and current Water Content amount, it is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
11.	TRAFO 4 UNIT 3	0071084208	14

A) Analysis

- **Acidity** of 0.020mgKOH/gm is **ACCEPTABLE**.
- **Dielectric** of 55kV is **ACCEPTABLE**.
- **Colour** of 1.0 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 17ppm is **ACCEPTABLE** while moisture in solid insulation of 2.01% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
12.	ASH HANDLING	10820108	15

A) Analysis

- **Acidity** of 0.022mgKOH/g is **ACCEPTABLE**.
- **Dielectric breakdown voltage** of 53kV is **ACCEPTABLE**.
- **Colour** of 1.0 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 16ppm is **ACCEPTABLE** while moisture in solid insulation of 1.89% M/dw is graded as **B**. For this transformer, it is recommended to maintain the water content less than 30ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
13.	ASH HANDLING	10820107	16

A) Analysis

- **Acidity** of 0.022mgKOH/g is **ACCEPTABLE**.
- **Dielectric breakdown voltage** of 31kV is **ACCEPTABLE**.
- **Colour** of 1.0 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 29ppm is assessed as **ACCEPTABLE**. Meanwhile, Water Content in solid insulation of 2.38% M/dw is graded as **C**. For this transformer, it is recommended to maintain the water content below 30ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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3.0 Summary of Our Comments & Recommendations:

No.	TX Unit No.	Serial No.	TC	Recommendation
1.	MAIN TRAFO UNIT 1	100691	4	• Periodic condition monitoring
2.	TRAFO 1 ESP UNIT 1	10937	5	• Perform OIL REGENERATION
3.	TRAFO 2 ESP UNIT 1	10035	6	• Perform OIL REGENERATION
4.	TRAFO 3 ESP UNIT 1	10031	7	• Periodic condition monitoring
5.	TRAFO 4 ESP UNIT 1	10034	8	• Perform OIL REGENERATION
6.	MAIN TRAFO UNIT 3	1122170	9	• Periodic condition monitoring
7.	UAT 3	1035082	10	• Periodic condition monitoring
8.	TRAFO 1 ESP UNIT 3	0073084208	11	• Perform OIL REGENERATION
9.	TRAFO 2 ESP UNIT 3	0070084208	12	• Perform OIL REGENERATION
10.	TRAFO 3 ESP UNIT 3	0072084208	13	• Perform OIL REGENERATION
11.	TRAFO 4 UNIT 3	0071084208	14	• Periodic condition monitoring
12.	ASH HANDLING	10820108	15	• Periodic condition monitoring
13.	ASH HANDLING	10820107	16	• Periodic condition monitoring

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